⑩ $\int_1^\infty \frac{1+\arcsin 1/x}{1+x\sqrt{x}}\,dx$ – единствена особеност при $x\to\infty$.

$\frac{1}{x}\xrightarrow[x\to\infty]{}0 \Rightarrow \arcsin\frac{1}{x}\xrightarrow[x\to\infty]{}0 \Rightarrow 1+\arcsin 1/x\sim 1,\ 1+x\sqrt{x}\underset{x\to\infty}{\sim}x\sqrt{x}=x^{3/2}$

$\Rightarrow \int_1^\infty \sim \int_1^\infty \frac{dx}{x^{3/2}} \to$ сходящ – сравнение с $\int_a^\infty \frac{dx}{a^x x^\alpha \ln^\beta x}$ – сх. при $\alpha>1$

⑪ $\int_0^\infty \frac{\ln(1+x^5)}{\sqrt{x+\sqrt{x}}}\,dx$ – особености $x=0$ и ∞. Тогава $\int_0^\infty = \int_0^1 + \int_1^\infty$

$\int_0^1 \to \ln(1+x^5)\sim x^5,\ \sqrt{x+\sqrt{x}}\sim x^{1/4} \Rightarrow \int_0^1 \sim \int_0^1 \frac{x^5}{x^{1/4}}\,dx = \int_0^1 x^{5-1/4}\,dx$ – сходящ

$\int_1^\infty \to 1+x^5\sim x^5 \Rightarrow \ln(1+x^5)\sim \ln x^5 = 5\ln x\sim \ln x,\ \sqrt{x+\sqrt{x}}\sim\sqrt{x}=x^{1/2}\Rightarrow$

$\int_1^\infty \sim \int_1^\infty \frac{\ln x}{\sqrt{x}}\,dx = \int_1^\infty \frac{dx}{x^{1/2}\ln^{-1}x}$ – разходящ – сравнение с $\int^\infty \frac{dx}{x^\alpha \ln^\beta x}$ - разх. при $\alpha<1$.

⑫ $\int_0^\infty \operatorname{arctg}\frac{x^3}{1+x^2}\cdot\frac{dx}{x^2}$ – особеност при $x=0$ и ∞. $\Rightarrow \int_0^\infty = \int_0^1 + \int_1^\infty$

$\int_0^1 \to \frac{x^3}{1+x^2}\underset{x\to0}{\sim}x^3 \Rightarrow \operatorname{arctg}\frac{x^3}{1+x^2}\underset{x\to0}{\sim}x^3 \Rightarrow \int_0^1 \sim \int_0^1 \frac{x^3}{x^2}\,dx = \int_0^1 x\,dx$ – сходящ

$\int_1^\infty \to \frac{x^3}{1+x^2}\xrightarrow[x\to\infty]{}\infty \Rightarrow \operatorname{arctg}\frac{x^3}{1+x^2}\xrightarrow[x\to\infty]{}\pi/2 \Rightarrow \operatorname{arctg}\frac{x^3}{1+x^2}\underset{x\to\infty}{\sim}1 \Rightarrow$

$\int_1^\infty \sim \int_1^\infty \frac{dx}{x^2}$ – сходящ. Окончателно $\int_0^\infty$ – сходящ.

⑬ $\int_3^\infty \frac{e^{-x}-\ln x}{(1+x^\alpha)^{\alpha-2}}\,dx$ – особеност при $x\to\infty$. $\lim_{x\to\infty} e^{-x}=0 \Rightarrow e^{-x}-\ln x \underset{x\to\infty}{\sim}\ln x$

$\Rightarrow \int_3^\infty \sim \int_3^\infty \frac{\ln x}{(1+x^\alpha)^{\alpha-2}}\,dx$

1) $\alpha<0 \to x^\alpha \xrightarrow[x\to\infty]{} 0 \Rightarrow 1+x^\alpha\sim 1 \Rightarrow (1+x^\alpha)^{\alpha-2}\sim 1 \Rightarrow \int_3^\infty \sim \int_3^\infty \frac{\ln x}{1}\,dx$ – разх.

2) $\alpha=0 \to 1+x^\alpha = 2 \Rightarrow 1+x^\alpha\sim 1 \to$ аналогично на 1)

3) $\alpha>0 \to 1+x^\alpha \sim x^\alpha \Rightarrow (1+x^\alpha)^{\alpha-2}\sim x^{\alpha(\alpha-2)} \Rightarrow \int_3^\infty \sim \int_3^\infty \frac{dx}{x^{\alpha(\alpha-2)}\ln^{-1}x}$

$\left.\begin{array}{l} \alpha(\alpha-2)>1 \to \text{сходящ} \\ \alpha(\alpha-2)<1 \to \text{разходящ} \\ \alpha(\alpha-2)=1 \to \text{разходящ} \end{array}\right\}$ квадратно неравенство.....

⑭ $\int_0^\infty \frac{\operatorname{arctg}^\alpha x}{(x^2+2)(e^x-1)^\beta}\,dx$ – особеност в 0 и $\infty \Rightarrow \int_0^\infty = \int_0^1 + \int_1^\infty$

$\int_0^1 \to \operatorname{arctg}x \underset{x\to0}{\sim} x,\ x^2+2 \underset{x\to0}{\sim} 1,\ e^x-1\sim x \Rightarrow \int_0^1 \sim \int_0^1 \frac{x^\alpha}{x^\beta}\,dx = \int_0^1 \frac{dx}{x^{\beta-\alpha}}$ – сходящ при $\beta-\alpha<1$.

$\int_1^\infty \to \lim_{x\to\infty}\operatorname{arctg}x = \frac{\pi}{2} \Rightarrow \operatorname{arctg}^\alpha x\sim 1,\ x^2+2\sim x^2,\ e^x-1\sim e^x \Rightarrow$

$\int_1^\infty \sim \int_1^\infty \frac{dx}{x^2 e^{\beta x}} \to \left\{\begin{array}{l} \beta>0 \to e^\beta > e^0 = 1 \Rightarrow \text{сходящ – сравнение с } \int_1^\infty \frac{dx}{a^x x^\alpha \ln^\beta x},\ a>1 \to \text{сх.} \\ \beta=0 \to \text{сходящ – сравнение с } \int^\infty \frac{dx}{x^\alpha},\ \text{сх. при } \alpha>1 \\ \beta<0 \to e^\beta<1 \text{ – разходящ –} \end{array}\right.$

Окончателно $\int_0^\infty$ е сходящ при $\left|\begin{array}{l} \beta-\alpha<1 \\ \beta\geq 0 \end{array}\right.$.

⑮ $\int_0^\infty \frac{\ln^\alpha(e^x-x)}{(x+\sqrt{x})^\beta \arcsin\frac{x}{x+1}}\,dx$ – особеност в 0 и $\infty \Rightarrow \int_0^\infty = \int_0^1 + \int_1^\infty$.

$\int_0^1 \to e^x = 1+x+\frac{x^2}{2}+\dots \Rightarrow e^x-x = 1+\frac{x^2}{2}+\dots \Rightarrow \ln(e^x-x) = \ln\left(1+\frac{x^2}{2}+\dots\right)\sim x^2$

$(x+\sqrt{x})^\beta \sim (\sqrt{x})^\beta = x^{\beta/2},\ \frac{x}{x+1}\xrightarrow[x\to0]{}0 \Rightarrow \arcsin\frac{x}{x+1}\sim\frac{x}{x+1}\sim x \Rightarrow$

$\int_0^1 \sim \int_0^1 \frac{x^{2\alpha}}{x^{\beta/2}\cdot x}\,dx = \int_0^1 \frac{dx}{x^{1+\beta/2-2\alpha}}$ – сходящ при $1+\frac{\beta}{2}-2\alpha<1$, т.е. $\frac{\beta}{2}-2\alpha<0 \Leftrightarrow \beta<4\alpha$.

$\int_1^\infty \to e^x-x \underset{x\to\infty}{\sim} e^x \Rightarrow \ln(e^x-x)\sim \ln e^x = x \Rightarrow \ln^\alpha(e^x-x)\sim x^\alpha$

$(x+\sqrt{x})^\beta \sim x^\beta,\ \frac{x}{x+1}\xrightarrow[x\to\infty]{}1 \Rightarrow \arcsin\frac{x}{x+1}\to \arcsin 1\sim 1 \Rightarrow$

$\int_1^\infty \sim \int_1^\infty \frac{x^\alpha}{x^\beta}\,dx = \int_1^\infty \frac{dx}{x^{\beta-\alpha}} \to$ сх. при $\beta-\alpha>1$

Окончателно $\int_0^\infty$ е сх. при $\left|\begin{array}{l} \beta<4\alpha \\ \beta-\alpha>1 \end{array}\right.$